

Investigating the Effect of Preprocessing, Model Complexity, and Regularisation on Generalisation in Epileptic Seizure Prediction

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Abstract—This report investigates how preprocessing choices, model complexity, regularisation strategies, and imbalance-handling methods affect generalisation performance in epileptic seizure prediction. Three EEG-related datasets were studied: the CHB-MIT Scalp EEG dataset, the Bangalore EEG Epilepsy Dataset (BEED), and the Epileptic Seizure Recognition dataset. Logistic regression was used as the baseline classifier so that differences in performance could be linked mainly to preprocessing and regularisation rather than model complexity. Two preprocessing pipelines were compared: Pipeline A used normalization, noise removal, and feature selection, while Pipeline B used feature extraction or feature selection followed by scaling and PCA. The results show that preprocessing order strongly affects performance, especially for structured EEG feature datasets. L1 regularisation generally provided the best balance between F1-score, sparsity, and generalisation, while Elastic Net did not consistently outperform L1 or L2. The study also shows that oversampling improves minority-class learning but can introduce synthetic noise, making sparse regularisation useful after imbalance correction.

Index Terms—Epileptic seizure prediction, EEG, preprocessing, logistic regression, L1 regularisation, L2 regularisation, Elastic Net, SMOTE, PCA, class imbalance.

I. INTRODUCTION

Epileptic seizure prediction is a difficult classification task because EEG data is noisy, non-stationary, high-dimensional, and often highly imbalanced. In real medical datasets, seizure events are much less frequent than non-seizure periods, so a model can achieve misleadingly high accuracy by predicting the majority class [4]. For this reason, F1-score and precision-recall AUC (PR-AUC) are more informative than accuracy alone.

The aim of this assignment is to study how preprocessing choices and regularisation strategies affect the ability of a logistic regression model to generalise across different seizure prediction datasets. Logistic regression is deliberately used as a simple baseline model. This makes it easier to observe whether performance changes are caused by the preprocessing pipeline, imbalance handling, or regularisation method.

The logistic regression model estimates the probability of seizure class as:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}} \quad (1)$$

The regularised objective can be written as:

$$J(W, b) = \frac{1}{m} \sum_{i=1}^m L(\hat{y}^{(i)}, y^{(i)}) + \frac{\lambda}{2m} \sum ||W||^2 \quad (2)$$

where the penalty term changes depending on the regularisation type:

- L1 regularisation penalises the absolute value of coefficients and can set some coefficients to zero.
- L2 regularisation penalises squared coefficient magnitude and keeps all features active.
- Elastic Net combines L1 and L2 penalties.

II. DATASET COLLECTION AND JUSTIFICATION

Three datasets were selected because they represent different EEG data characteristics: raw continuous EEG, structured multichannel EEG features, and high-dimensional flattened time-series EEG segments (see Table I).

A. Dataset 1: CHB-MIT Scalp EEG Dataset

The CHB-MIT Scalp EEG dataset [1] contains EEG recordings from pediatric epilepsy patients. The dataset is large and continuous, making it suitable for realistic seizure prediction experiments.

Key properties:

- Total dataset size: 42.62 GB
- Total EDF files in the Kaggle-prepared experiment: 686
- Files with seizures: 141
- Files without seizures: 545
- Seizure-file percentage: 20.55%
- Total recorded seizures in the prepared dataset notes: 198
- Total recording duration: approximately 1000 hours

The dataset is highly imbalanced because most EEG time belongs to normal or interictal activity. Depending on segmentation, seizure to non-seizure ratios may reach approximately 1:100 to 1:500. This makes it useful for testing whether SMOTE, feature selection, and regularisation can improve minority-class detection.

Feature characteristics:

- Raw EEG time-series data

TABLE I
DATASET COLLECTION AND JUSTIFICATION

Dataset	Source	Data Type	Size Used / Reported	Main Imbalance	Justification
CHB-MIT Scalp EEG	PhysioNet/Kaggle copy	Raw pediatric scalp EEG, EDF files	42.62 GB, 686 EDF files	141 seizure files vs 545 non-seizure files	Real-world, noisy, continuous EEG with severe imbalance
BEED	UCI Machine Learning Rep.	Tabular multichannel EEG features	8000 instances, 16 EEG channels	3:1 binary imbalance before balancing	Clean, structured, low-dimensional EEG feature dataset
Epileptic Seizure Rec.	Kaggle/UCI-derived dataset	Flattened EEG time-series values	11500 samples, 178 features	9200 non-seizure vs 2300 seizure	High-dimensional 1-second EEG segments with clear binary conversion

- High-dimensional, multichannel, continuous recordings
- Non-stationary patterns that vary across patients
- High noise from eye movement, muscle activity, and recording artifacts
- Feature engineering required before classification

For computational feasibility, the notebook used a compact file-level feature representation for initial logistic regression experiments. The selected feature count remained 10 after variance-threshold feature selection, so the CHB-MIT results should be interpreted as a preliminary baseline showing the difficulty of learning from limited engineered features.

B. Dataset 2: BEED - Bangalore EEG Epilepsy Dataset

The BEED dataset [2] was selected as a structured EEG dataset from the UCI Machine Learning Repository. Unlike CHB-MIT, it is already stored in a tabular format and contains 16 EEG channel features.

Key properties:

- Total instances used: 8000
- EEG channel features: 16, labelled X1-X16
- Sampling rate: 256 Hz
- Segment length: 20 seconds
- Subjects: 80 individuals
- No missing values

In the notebook implementation, the original labels were converted to a binary target. The resulting class distribution before balancing was 6000 samples in the majority class and 2000 samples in the minority class, giving an imbalance ratio of approximately 3:1. Borderline-SMOTE was then applied to balance the classes to 6000 samples each. This dataset is cleaner than CHB-MIT because it is already feature-engineered and low-dimensional. Therefore, it is useful for testing whether preprocessing order matters even when the features are already meaningful.

C. Dataset 3: Epileptic Seizure Recognition Dataset

The Epileptic Seizure Recognition dataset [3] contains 11500 EEG segments. Each sample contains 178 time-series values representing a one-second EEG segment. The original dataset contains five classes, but it was converted into a binary seizure detection problem.

Key properties:

- Total samples: 11500
- Features: 178
- Original classes: 5
- Binary target: Seizure (class 1) and Non-seizure (classes 2, 3, 4, and 5)
- EEG segment length: 1 second
- Sampling points per segment: 178

Class distribution before balancing:

- **Non-seizure (Class 0):** 9200
- **Seizure (Class 1):** 2300

The imbalance ratio is approximately 4:1. Oversampling was applied to balance the dataset to 9200 non-seizure and 9200 seizure samples. This dataset is higher-dimensional than BEED but more structured than CHB-MIT. It is useful for studying how PCA, feature selection, and sparse regularisation behave when the feature space is larger.

III. PREPROCESSING PIPELINES

Two main preprocessing pipelines were compared across the datasets.

A. Pipeline A: Normalization $\rightarrow$ Noise Removal $\rightarrow$ Feature Selection

Pipeline A applies scaling and cleaning before feature selection. The goal is to preserve discriminative EEG features while removing noisy or irrelevant dimensions.

Steps:

- 1) Normalize or standardize features.
- 2) Remove noise or outliers using filtering or Isolation Forest.
- 3) Select the strongest features using variance thresholding or statistical feature selection.
- 4) Train logistic regression.

Pipeline A is more interpretable because it keeps selected original features instead of mixing them into PCA components.

B. Pipeline B: Feature Extraction / Selection → Scaling → PCA

Pipeline B applies dimensionality reduction after scaling. The aim is to compress the feature space and reduce redundancy.

Steps:

- 1) Extract or select features.
- 2) Scale features.
- 3) Apply PCA.
- 4) Train logistic regression.

Pipeline B can reduce noise, but PCA may also mix important seizure-specific signals with non-discriminative variance. Therefore, PCA can preserve total variance while still losing class-discriminative information.

IV. BASELINE LOGISTIC REGRESSION RESULTS

A. Dataset 1: CHB-MIT Baseline Results

Pipeline B slightly outperformed Pipeline A for CHB-MIT (Table II). This suggests that transforming raw EEG into a compact feature space can help more than simple cleaning alone. However, both scores remain low, which confirms that raw continuous EEG requires stronger feature engineering than simple file-level features.

B. Dataset 2: BEED Baseline Results

Pipeline A performed much better than Pipeline B in F1-score (Table III). Although Pipeline B retained 98.67% explained variance, the F1-score dropped from 0.7094 to 0.5881. This shows that PCA preserved variance but lost class-discriminative information.

C. Dataset 3: Epileptic Seizure Recognition Baseline Results

Pipeline A and Pipeline B produced similar F1-scores, but with different trade-offs (Table IV). Pipeline A had higher accuracy and stronger ranking ability, while Pipeline B gave a more balanced but less stable classifier.

V. OVERFITTING AND UNDERFITTING DEMONSTRATION

The assignment required intentional creation of underfitting and overfitting scenarios. These were implemented using learning curves in the notebooks.

A. Underfitting Scenario

Underfitting was created using:

- Very strong regularisation, such as very small C values.
- A limited number of selected features, for example top 2 features only.
- Simple logistic regression with insufficient capacity to capture EEG patterns.

Expected learning-curve behavior: Training score is low. Validation score is also low. The gap between training and validation is small. Both curves converge at a poor score.

Interpretation: The model has high bias. It is too simple and cannot learn meaningful seizure patterns.

B. Overfitting Scenario

Overfitting was created using:

- No regularisation or very weak regularisation.
- A larger feature space, such as all original features.
- High-dimensional input relative to useful signal.

Expected learning-curve behavior: Training score is higher than validation score. Validation performance remains lower. A visible gap appears between training and validation curves.

Interpretation: The model has high variance. It fits noise and dataset-specific patterns rather than general seizure characteristics.

C. Regularisation Tuning Result

For BEED, hyperparameter tuning with L1 regularisation found the optimal parameters outlined in Table V. This shows that moderate-to-strong regularisation improved generalisation and reduced overfitting.

VI. REGULARISATION STUDY

Three regularisation methods were compared: L1, L2, and Elastic Net.

A. CHB-MIT Regularisation Results

L1 performed best on CHB-MIT (Table VI) because it removed irrelevant features from a noisy feature representation. L2 kept all features active and performed worst, showing that retaining noisy features can damage generalisation. Stability across 10 random splits is shown in Table VII. L1 had the highest mean F1-score and the lowest F1 variation, so it generalised best in this experiment.

B. BEED Regularisation Results

All regularisation methods performed similarly on BEED (Table VIII). L1 gave the best F1-score, while L2 gave the best PR-AUC. This happens because BEED is cleaner and lower-dimensional, so there is less need for aggressive sparsity.

C. Epileptic Seizure Recognition Regularisation Results

All three regularisation methods produced very similar performance (Table IX), but L1 and Elastic Net produced useful sparsity. L1 removed 40% of the selected features, while Elastic Net removed 35%. This suggests that the dataset contains redundant or noisy features even after feature selection.

TABLE II
CHB-MIT BASELINE RESULTS

Pipeline	Feature Space	Accuracy	F1-score	PR-AUC	Interpretation
A	10 selected features	0.4404	0.4404	0.4703	Weak learning from limited engineered features
B	6 extracted features reduced to 4 PCA components	0.4771	0.4771	0.4915	Slight improvement after feature extraction and PCA

TABLE III
BEED BASELINE RESULTS

Pipeline	Transformation	Accuracy	F1-score	PR-AUC	Interpretation
A	16 features $\rightarrow$ 10 selected features; 11400 samples after noise removal	0.6579	0.7094	0.5807	Best F1-score; strong balance between precision and recall
B	16 features $\rightarrow$ 10 PCA components; 98.67% variance retained	~ 0.59	0.5881	0.5927	Higher PR-AUC but much lower F1-score

TABLE IV
EPILEPTIC SEIZURE RECOGNITION BASELINE RESULTS

Pipeline	Transformation	Accuracy	F1-score	PR-AUC	Interpretation
A	178 features $\rightarrow$ 20 selected features; 17480 samples after noise removal	0.6519	0.5321	0.6616	Higher accuracy and PR-AUC, but low seizure recall
B	178 features $\rightarrow$ 20 PCA components; 71.86% variance retained	~ 0.57	0.5351	0.6689	Similar F1-score but weaker overall accuracy

TABLE V
BEED HYPERPARAMETER TUNING

Best Parameter	Best Cross-Validation F1-score
C = 0.01	0.7397

TABLE VI
CHB-MIT REGULARISATION PERFORMANCE

Model	Accuracy	F1-score	Sparsity
L1 (Lasso)	0.5229	0.5273	50%
L2 (Ridge)	0.4404	0.4404	0%
Elastic Net	0.4679	0.4957	30%

TABLE VII
CHB-MIT STABILITY ACROSS 10 RANDOM SPLITS

Model	Mean F1	F1 Std	Mean Sparsity	Sparsity Std
L1 (Lasso)	0.5392	0.0276	51%	15.95
L2 (Ridge)	0.4866	0.0451	0%	0.00
Elastic Net	0.4854	0.1028	43%	20.03

D. Sparsity Comparison

The sparsity comparison (Table X) shows that higher-dimensional EEG data requires more feature elimination to

TABLE VIII
BEED REGULARISATION RESULTS

Model	F1-score	PR-AUC	Zero Coefficients
L1 (Lasso)	0.7247	0.569113	3 out of 10
L2 (Ridge)	0.7165	0.576886	0 out of 10
Elastic Net	0.7216	0.572908	2 out of 10

TABLE IX
EPILEPTIC SEIZURE RECOGNITION REGULARISATION RESULTS

Model	F1-score	PR-AUC	Zero Coefficients
L1 (Lasso)	0.5247	0.667060	8 out of 20
L2 (Ridge)	0.5248	0.666577	0 out of 20
Elastic Net	0.5215	0.666311	7 out of 20

generalise well.

VII. HANDLING CLASS IMBALANCE

Class imbalance was handled mainly using oversampling-based techniques.

A. Techniques Used

For future improvement, class weighting or random under-sampling can be added as a second non-oversampling baseline. However, the current experiments already show the important

TABLE X
SPARSITY COMPARISON ACROSS DATASETS

Dataset	Active Features Used	L1 Zeros	Elastic Net Zeros	Interpretation
BEED	10	3	2	Low-dimensional structured data needs moderate sparsity
Epileptic Seizure Rec.	20	8	7	Higher-dimensional data needs stronger sparsity

TABLE XI
IMBALANCE HANDLING TECHNIQUES USED

Technique	Used On	Purpose	Main Effect
Borderline-SMOTE	CHB-MIT and BEED notebook experiments	Generate synthetic samples near the decision boundary	Improves minority-class learning but may introduce synthetic noise
Oversampling / SMOTE-style balancing	Epileptic Seizure Recognition dataset	Balance seizure and non-seizure samples	Improves recall opportunity for seizure class
Regularised logistic regression after balancing	All datasets	Reduce noise introduced by oversampling	L1 and Elastic Net remove redundant synthetic patterns

interaction between synthetic balancing and regularisation (Table XI).

B. Class Imbalance and Sparsity Interaction

For CHB-MIT, the L1 sparsity changed as the sampling ratio increased:

TABLE XII
CHB-MIT SPARSITY VS SAMPLING RATIO

Sampling Ratio	L1 Sparsity
0.3	70%
0.5	50%
0.8	30%
1.0	40%

Interpretation: At low sampling ratios, the dataset remains strongly imbalanced, so the model uses only the strongest features and removes many coefficients. As the sampling ratio increases, more seizure-like patterns become available, so fewer features are removed. At full balance, sparsity increases slightly again, probably because SMOTE introduces synthetic noise and the L1 penalty removes redundant features.

Final insight: There is a strong interaction between class imbalance and regularisation. As the dataset becomes more balanced, the model relies on a broader set of features, reducing sparsity. However, excessive synthetic data may introduce noise, requiring feature elimination again.

VIII. COMPARATIVE ANALYSIS

A. Does Preprocessing Order Affect Results?

Yes. Preprocessing order strongly affects results. Across the datasets, Pipeline A was generally more stable and more interpretable. Pipeline A clearly outperformed Pipeline B on

BEED, where F1-score dropped from 0.7094 to 0.5881 after PCA. This shows that PCA can preserve variance while losing class-discriminative information.

For the Epileptic Seizure Recognition dataset, Pipeline A and B produced similar F1-scores, but Pipeline A had better accuracy and PR-AUC. For CHB-MIT, Pipeline B slightly improved performance because feature extraction and PCA helped convert raw EEG into a more useful compact representation.

EEG signals contain subtle seizure-related patterns that are often better preserved by feature selection than by PCA compression. PCA is useful for reducing redundancy, but it can dilute medically relevant patterns.

B. Which Regularisation Generalises Best Across Datasets?

L1 regularisation generalised best overall. **Reasons:**

- It achieved the best F1-score on CHB-MIT.
- It achieved the best F1-score on BEED.
- It produced useful sparsity on the Epileptic Seizure Recognition dataset.
- It removed noisy or redundant features after oversampling.

L2 was more stable in some PR-AUC results because it keeps all features, but it did not provide feature selection. Elastic Net was competitive but did not consistently outperform L1.

C. Does Elastic Net Consistently Outperform L1 or L2?

No. Elastic Net did not consistently outperform L1 or L2. Elastic Net usually performed between L1 and L2. It helped with sparsity, but it did not provide a clear advantage over L1 in these datasets. This suggests that many EEG features were either useful or redundant, making pure L1 feature elimination more effective than a mixed penalty.

D. How Does Imbalance Handling Interact With Regularisation?

Imbalance handling and regularisation interact strongly. SMOTE and Borderline-SMOTE improve the representation of seizure samples, but they can also introduce synthetic noise. L1 and Elastic Net regularisation act as filters after oversampling by shrinking or removing weak features. This effect was strongest in noisy or higher-dimensional data. For CHB-MIT, strong imbalance caused high L1 sparsity. As the data became more balanced, the model used more features. At full balance, sparsity increased again, suggesting that synthetic samples introduced some redundant information.

IX. FINAL CROSS-DATASET COMPARISON

X. CONCLUSION

This study shows that preprocessing order, imbalance handling, and regularisation significantly affect seizure prediction performance. Pipeline A was generally stronger because it preserved interpretable EEG features and removed irrelevant features directly. Pipeline B sometimes helped reduce noise, but PCA often reduced F1-score because it mixed features and weakened class-discriminative patterns.

L1 regularisation was the most useful regularisation method overall because it improved F1-score in most cases and produced sparsity. L2 was stable but kept noisy features, while Elastic Net did not consistently outperform L1 or L2. The results also show that oversampling improves seizure detection opportunity, but synthetic samples can introduce noise, making sparse regularisation important after imbalance correction.

The effectiveness of preprocessing and regularisation in seizure prediction is highly dependent on data characteristics. Raw EEG data requires aggressive feature selection and sparsity, while structured datasets benefit more from careful preprocessing order. Across all datasets, L1 regularisation consistently provides the best balance between performance and generalisation by filtering noise and redundant features.

XI. RECOMMENDED FIGURES FOR FINAL SUBMISSION

The notebooks already generate visualisations that should be inserted into the final Word/PDF submission (see Table ??).

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TABLE XIII
FINAL CROSS-DATASET COMPARISON

Aspect	CHB-MIT	BEED	Epileptic Seizure Recognition
Data type	Raw continuous EEG	Structured tabular EEG features	Flattened EEG time-series segments
Dimensionality	High in original form, compact in experiment	Low, 16 channels	High, 178 features
Noise level	High	Low to moderate	Moderate
Imbalance	Severe	Moderate	4:1
Best pipeline	Pipeline B slightly in preliminary feature setup	Pipeline A	Pipeline A by accuracy/PR-AUC; Pipeline B similar F1
Best regularisation	L1	L1 by F1, L2 by PR-AUC	L1/L2 nearly tied, L1 more sparse
Main lesson	Raw EEG needs stronger feature engineering	Preserve meaningful features	Higher-dimensional data benefits from sparsity